

Mikroişlemci Sistemleri

Dr. Öğr. Üyesi Erkan Uslu

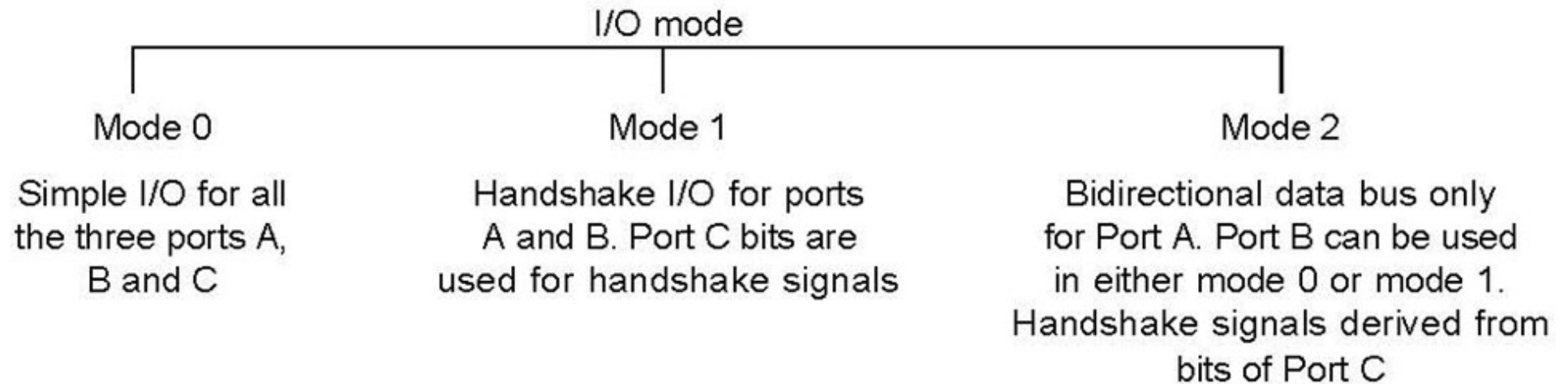
4

YTÜ-CE

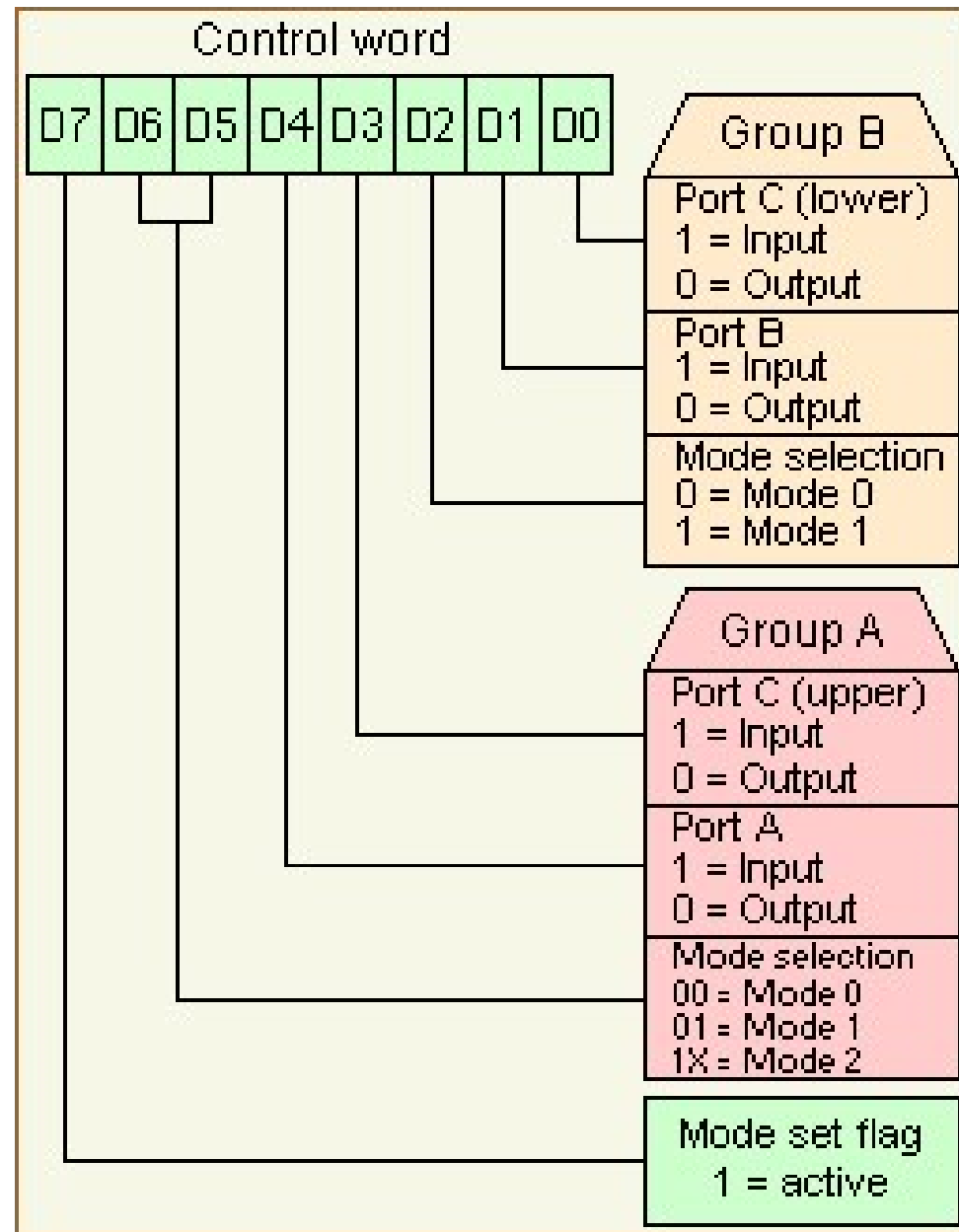
Ders-4 Konular

- 8255 modları
 - Mod 0
 - LED ve Buton
 - 7 parçalı gösterge
 - 3x4 tuş tarama
 - BSR (Bit set reset)
 - Mod 1
 - Mod 2

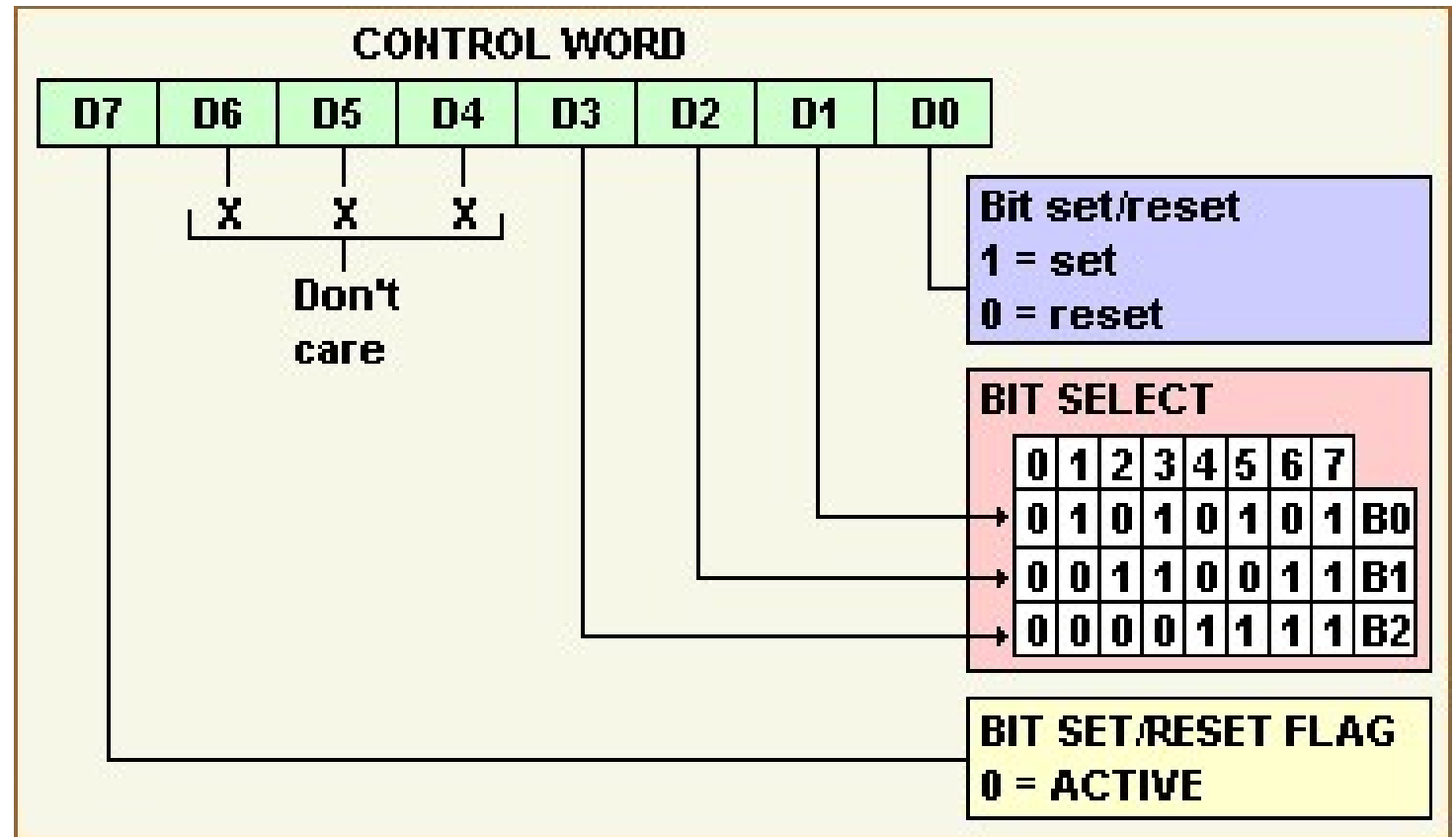
8255 Modlar



8255
Kontrol
Yazmacı →
Mod 0



8255
Kontrol
Yazmacı →
BSR



BSR Mod Örneği

- Örnek:
- 80H adresinden itibaren ardışık çift adreslere yerleştirilmiş bir 8255'de
 - PC2'yi lojik 1 olacak şekilde
 - PC6'da ise duty cycle'ı %66 olan bir kare dalga üretecek şekilde

programlayın

BSR Mod Örneği

- MOV AL, 00000101B
OUT 86H, AL
- AGAIN MOV AL, 0xxx1101
 OUT 86H, AL
 CALL Delay
 CALL Delay
 MOV AL, 0xxx1100
 OUT 86H, AL
 CALL Delay
 JMP AGAIN

8255 Mod 1

- Handshaking sinyalleri ile kontrollü tek yönlü veri gönderme veya almayı sağlar. (Mod 2'de handshaking sinyalleri ile çift yönlü veri iletimi)
- PORTA ve/veya PORTB Mod 1 için tek yönlü olarak veri iletimi için kullanılırken, PORTC uçları handshaking işaretleri için kullanılır.

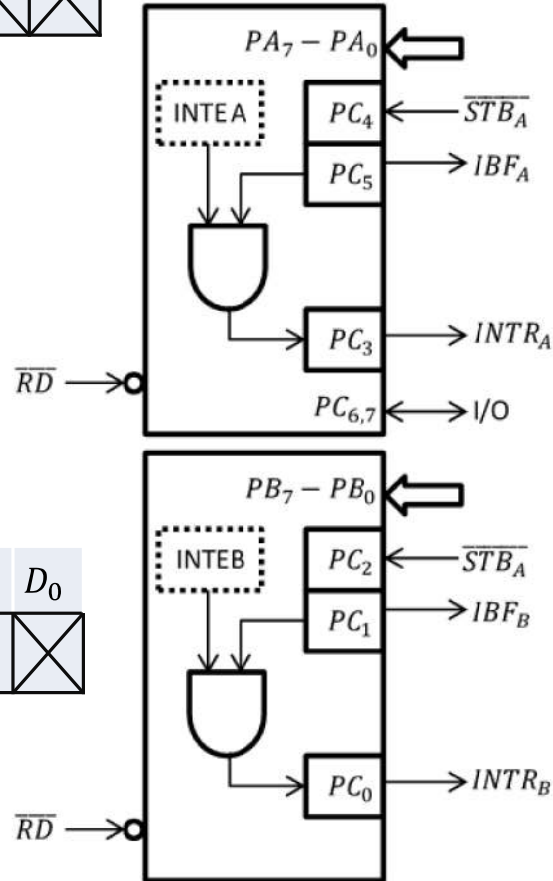
8255 Mod 1

- Grup A ve Grup B ayrı ayrı Mod 1 için programlanabilir.
- Her grupta 8 bit veri 4 bit kontrol işareti vardır.

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	0	1	1	1/0	X	X	X

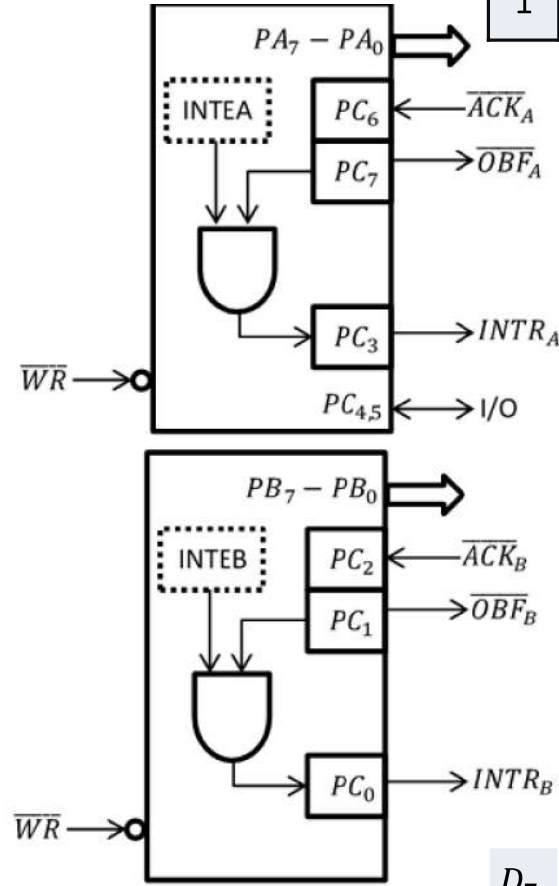
$PC_{6,7}$
1=Input
0=Output

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	X	X	X	X	1	1	X



D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	0	1	0	1/0	X	X	X

$PC_{4,5}$
1=Input
0=Output



D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	X	X	X	X	1	0	X

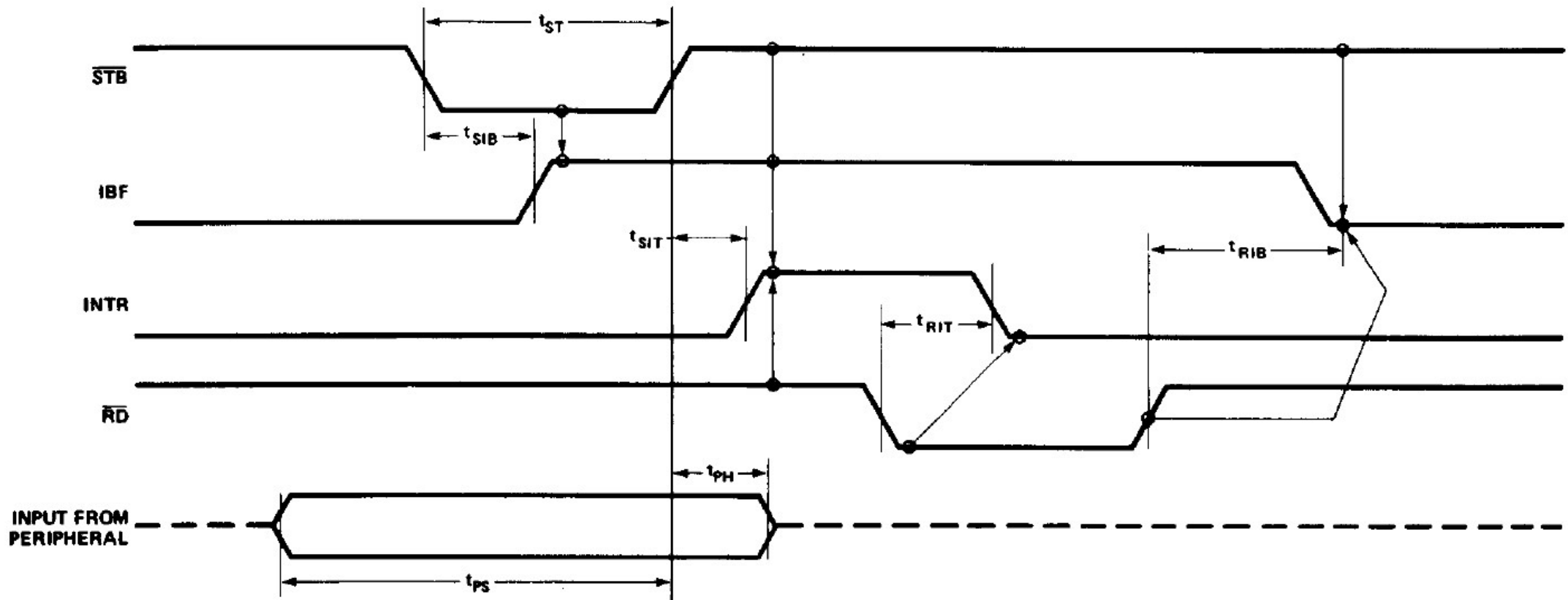
8255 Mod 1 - Input

- $\overline{STB}$: A “low” on this input loads data into the input latch
- IBF : A “high” on this output indicates that the data has been loaded into the input latch; in essence, an acknowledgement. IBF is set by $\overline{STB}$ input being low and is reset by the rising edge of the $\overline{RD}$ input

8255 Mod 1 - Input

- INTR : A “high” on this output can be used to interrupt the CPU when an input device is requesting service. INTR is set by the $\overline{STB}$ is a “1” IBF is a “1” and INTE is a “1”. It is reset by the falling edge of $\overline{RD}$
- INTEA : Controlled by bit set/reset of PC4
- INTEB : Controlled by bit set/reset of PC2

8255 Mod 1 – Input Timing



8255 Mod 1 - Output

- $\overline{OBF}$: goes “0” to indicate that the CPU has written data out to the specified port. The $\overline{OBF}$ will be set by the rising edge of the $\overline{WR}$ input and reset by $\overline{ACK}$ input being low
- $\overline{ACK}$: A “0” on this input informs the 82C55A that the data from Port A or Port B has been accepted

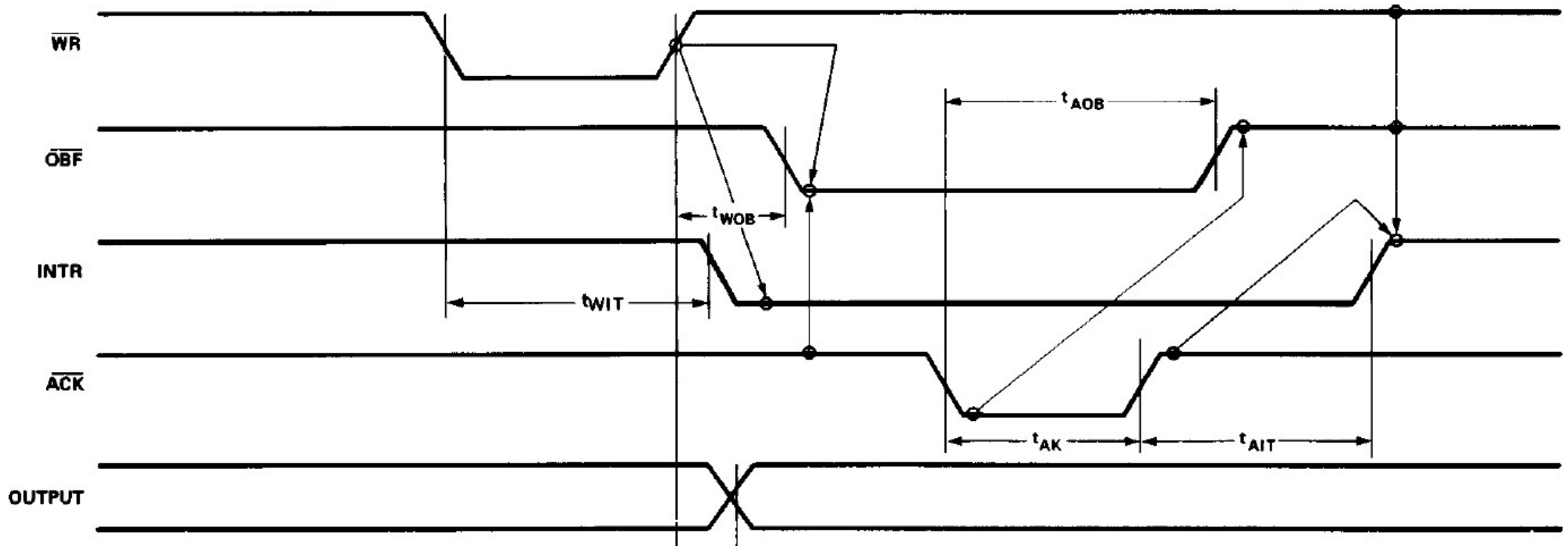
8255 Mod 1 - Output

- INTR : A “0” on this output can be used to interrupt the CPU when an output device has accepted data transmitted by the CPU. INTR is set when $\overline{ACK}$ is a “1”, $\overline{OBF}$ is a “1” and INTE is a “1”. It is reset by the falling edge of $\overline{WR}$.

8255 Mod 1 - Output

- INTEA : Controlled by bit set/reset of PC6
- INTEB : Controlled by bit set/reset of PC2

8255 Mod 1 – Output Timing



8255 Mod 1 – Status Word

- 8255 mod 1 için ayarlanmışsa PORTC'den yapılan okumalar STATUS WORD'dür
- OBF, IBF, INTR değerleri ile I/O için kullanılan PORTC uçları okunabilir

8255 Mod 1 – Status Word

- Input

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
I/O	I/O	IBF_A	$INTE_A$	$INTR_A$	$INTE_B$	IBF_B	$INTR_B$
GROUP A					GROUP B		

- Output

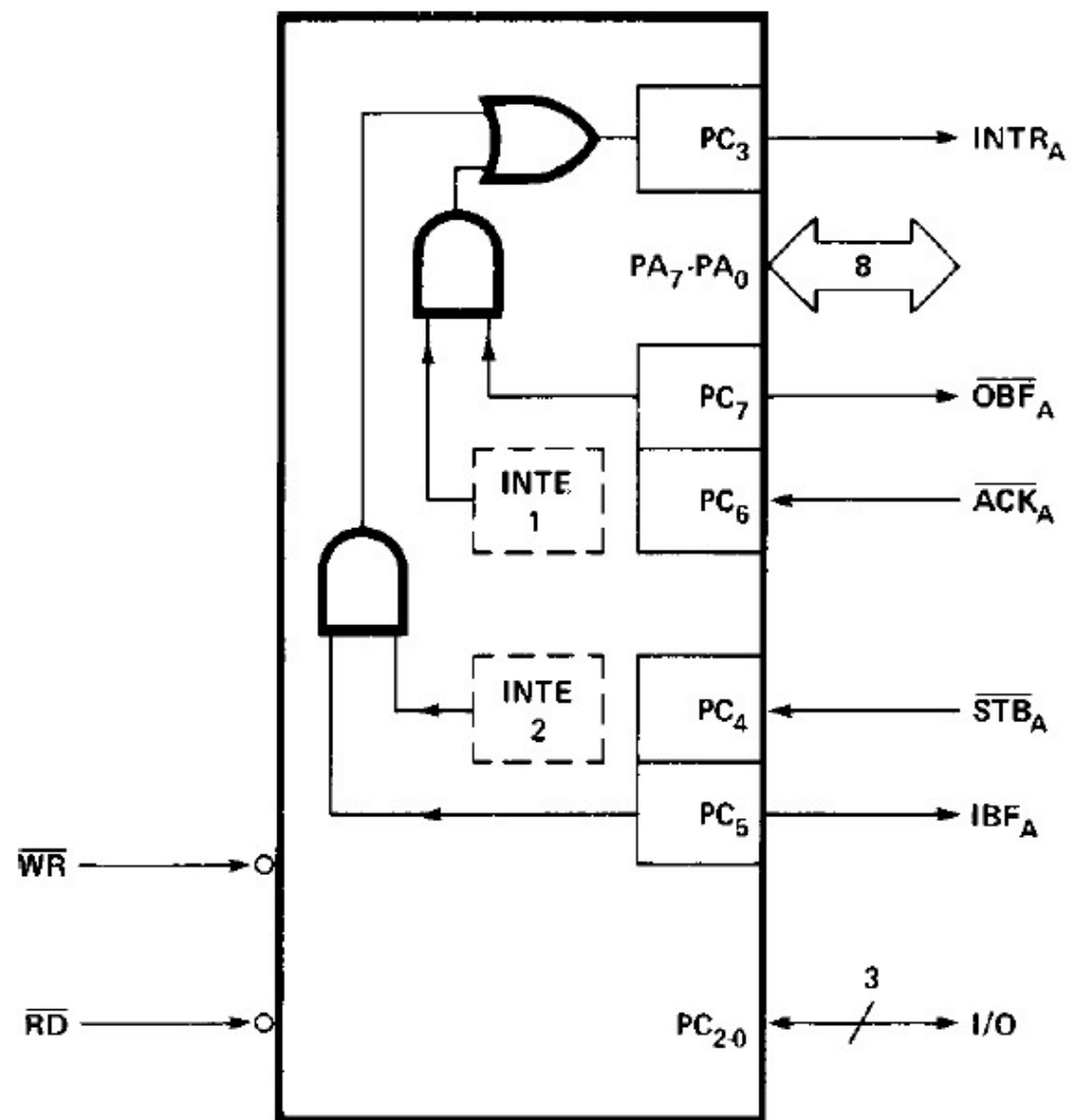
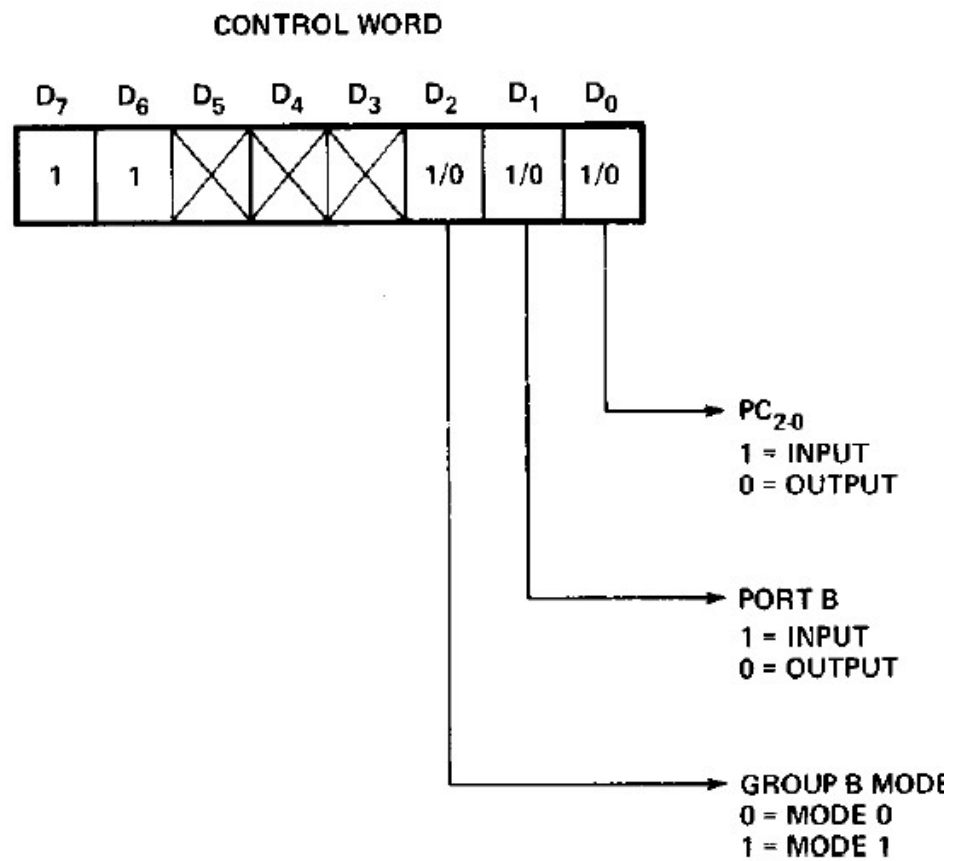
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
OBF_A	$INTE_A$	I/O	I/O	$INTR_A$	$INTE_B$	OBF_B	$INTR_B$
GROUP A					GROUP B		

8255 Mod 1

- 8255 Grup A Mod 1 output, 8255 Grup B Mod 1 input yönlü ayarlanarak, bunlar arasında handshaking tabanlı I/O işlemleri nasıl yapılabilir?

8255 Mod 2

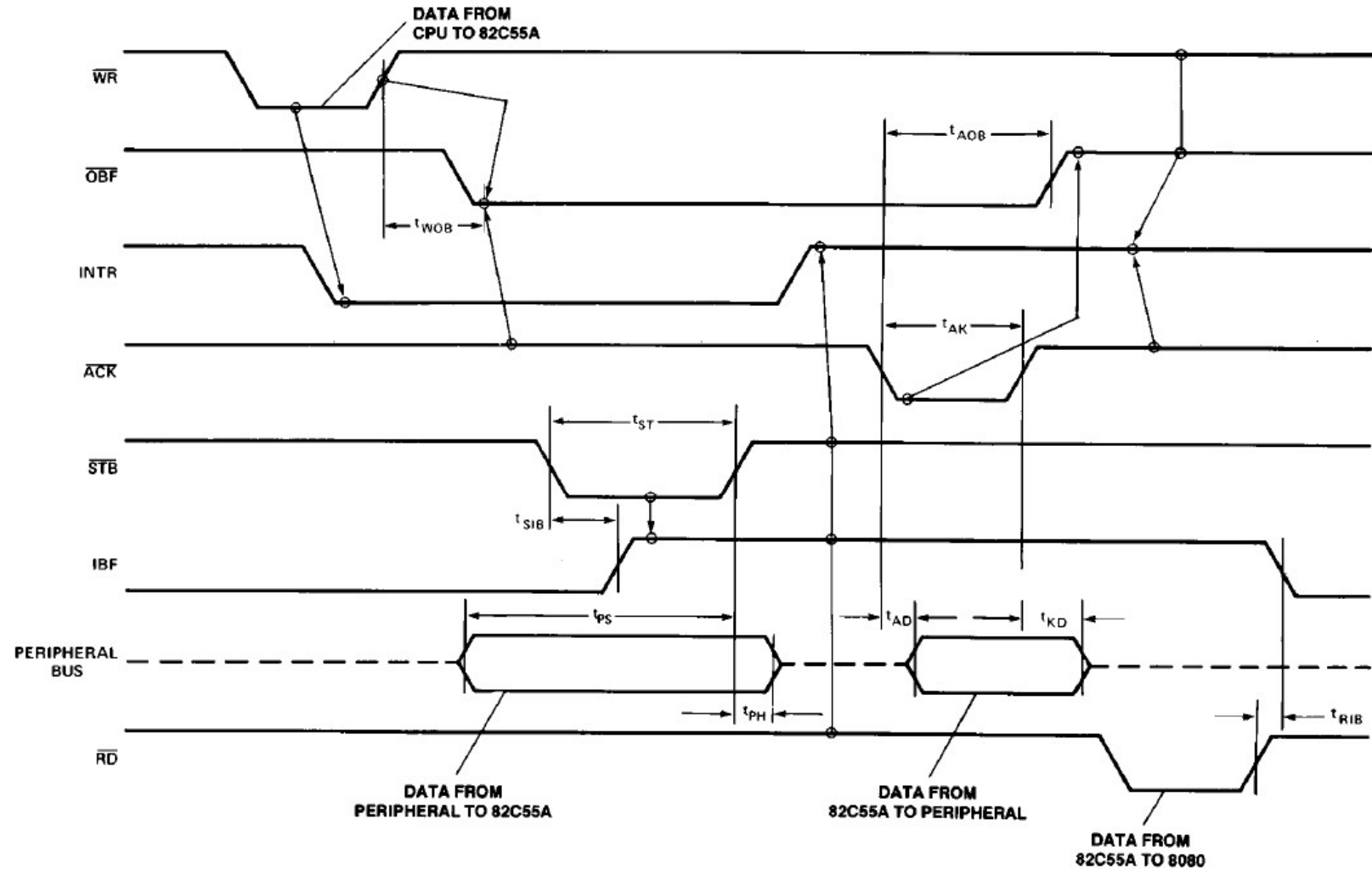
- Strobed bidirectional bus I/O
- Sadece Grup A mod 2 destekler
- 1 adet 8 bit çift yönlü port (PORTA), ve 5 handshaking kontrol işareti mevcuttur



8255 Mod 2

- INTE 1 : Controlled by bit set/reset of PC6
- INTE 2 : Controlled by bit set/reset of PC4

8255 Mod 2 - Timing



8255 Mod 2 – Status Word

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
OBF_A	$INTE_1$	IBF_A	$INTE_2$	$INTR_A$			
GROUP A					GROUP B		

8255

Modlar:

Özet

Tablo

	MODE 0	
	IN	OUT
PA ₀	IN	OUT
PA ₁	IN	OUT
PA ₂	IN	OUT
PA ₃	IN	OUT
PA ₄	IN	OUT
PA ₅	IN	OUT
PA ₆	IN	OUT
PA ₇	IN	OUT
PB ₀	IN	OUT
PB ₁	IN	OUT
PB ₂	IN	OUT
PB ₃	IN	OUT
PB ₄	IN	OUT
PB ₅	IN	OUT
PB ₆	IN	OUT
PB ₇	IN	OUT
PC ₀	IN	OUT
PC ₁	IN	OUT
PC ₂	IN	OUT
PC ₃	IN	OUT
PC ₄	IN	OUT
PC ₅	IN	OUT
PC ₆	IN	OUT
PC ₇	IN	OUT

MODE 1	
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
IN	OUT
INTR _B	INTR _B
IBF _B	OBF _B
STB _B	ACK _B
INTR _A	INTR _A
STB _A	I/O
IBF _A	I/O
I/O	ACK _A
I/O	OBF _A

MODE 2
GROUP A ONLY
↔
↔
↔
↔
↔
↔
↔
↔
—
—
—
—
—
—
—
I/O
I/O
I/O
INTR _A
STB _A
IBF _A
ACK _A
OBF _A

MODE 0
OR MODE 1
ONLY